Distance and Similarity Metrics, and the Silhouette Score

INT396 • Unsupervised Learning • Units I–II

School of Computer Science & Engineering

Student handout

Every clustering algorithm groups things that are *close* together. The algorithm itself has no notion of what "close" means — it receives a table of numbers. You supply the meaning, by choosing a distance metric.

That choice is not a preprocessing detail. With the same data and the same algorithm, changing the metric changes the answer. This handout works through the three metrics on your syllabus, explains when and why the choice matters, introduces the silhouette score for judging a clustering, and ends with numerical questions.

Contents

1 Distance and similarity

A **distance** function $d(p, q)$ returns a number that is small when two objects are alike. A **similarity** function returns a number that is large when two objects are alike. They point in opposite directions, and mixing them up reverses every conclusion that follows.

A function d is a genuine **metric** when it satisfies four conditions:

$$d(p,q) \geq 0 \qquad \text{non-negativity} \tag{1}$$
$$d(p,p) = 0 \qquad \text{identity} \tag{2}$$
$$d(p,q) = d(q,p) \qquad \text{symmetry} \tag{3}$$
$$d(p,r) \leq d(p,q) + d(q,r) \qquad \text{triangle inequality} \tag{4}$$

The fourth is obvious once stated in words: you cannot get there faster by going somewhere else first.

Throughout, an object is a vector of n measured features, $p = (p_1, p_2, \ldots, p_n)$.

2 Euclidean distance

The straight-line distance between two points — Pythagoras, generalised to any number of features.

$$d_{\text{euc}}(p,q) \;=\; \sqrt{\sum_{i=1}^{n} (p_i - q_i)^2} \tag{5}$$

where

n the number of features

p_i the value of feature i for object p

$(\ )^2$ squares each difference, which removes the sign *and* penalises one large gap far more than several small ones

$\sqrt{\ }$ returns the answer to the original units

As a recipe: subtract feature by feature, square each difference, add them all, take the square root.

Worked example 2.1 — $p = (2,5)$ *and* $q = (5,9)$

Subtract, feature by feature

$$p_1 - q_1 = 2 - 5 = -3, \qquad p_2 - q_2 = 5 - 9 = -4$$

Square each difference

$$(-3)^2 = 9, \qquad (-4)^2 = 16$$

Add, then take the square root

$$d_{\text{euc}}(p,q) = \sqrt{9+16} = \sqrt{25} = 5$$

Euclidean distance is the default choice when features are continuous, measured in comparable units, and straight-line closeness is physically meaningful.

3 Manhattan distance

The distance travelled when movement is restricted to the axes — as in a city laid out on a grid, where you cannot walk through the buildings.

$$d_{\mathrm{man}}(p, q) \;=\; \sum_{i=1}^{n} |\, p_i - q_i \,| \tag{6}$$

where

$|\cdot|$ absolute value: the size of the difference, without its sign

no square every feature contributes in proportion to its own difference, so one large gap does not dominate

---

Worked example 3.1 — *The same $p = (2, 5)$ and $q = (5, 9)$*

Take absolute differences and add them

$$d_{\mathrm{man}}(p, q) = |2 - 5| + |5 - 9| = 3 + 4 = 7$$

The crow flies 5; the pedestrian walks 7. Neither answer is wrong — they answer different questions.

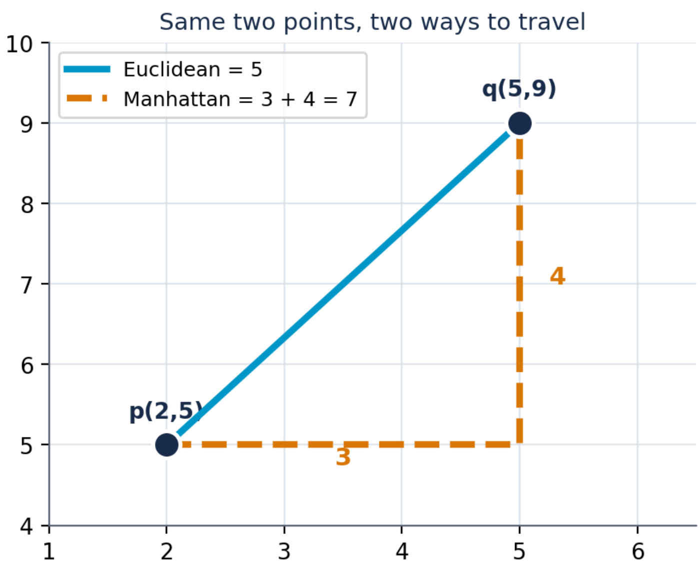

Figure 1: The same two points under the two metrics. The solid line is the Euclidean distance; the dashed path is the Manhattan distance.

A useful check. The straight line can never be longer than the path along the axes, so

$$d_{\mathrm{man}}(p, q) \;\geq\; d_{\mathrm{euc}}(p, q) \quad \text{always.}$$

If you ever compute a smaller Manhattan distance than Euclidean for the same pair, you have made an arithmetic error.

A consequence worth seeing. Consider the set of all points at distance exactly 1 from the origin. Under the Euclidean metric this set is a circle; under the Manhattan metric it is a diamond. “Nearby” is literally a different shape, which is why the two metrics produce different clusters.

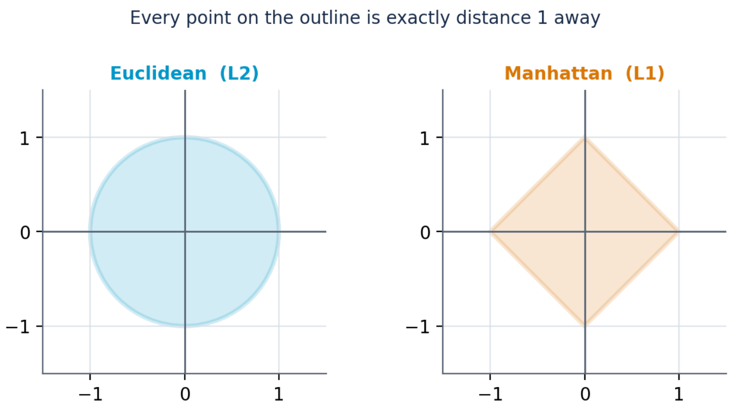

Figure 2: Every point on each outline is exactly distance 1 from the centre.

4 Cosine similarity

Cosine similarity measures the *angle* between two vectors and ignores their lengths entirely.

$$\cos(\theta) = \frac{\mathbf{A}\cdot\mathbf{B}}{\|\mathbf{A}\|\,\|\mathbf{B}\|} = \frac{\sum_{i=1}^{n} A_i B_i}{\sqrt{\sum_{i=1}^{n} A_i^2}\,\sqrt{\sum_{i=1}^{n} B_i^2}} \tag{7}$$

where

$\mathbf{A}\cdot\mathbf{B}$ the dot product: multiply matching components, then add

$\|\mathbf{A}\|$ the length of $\mathbf{A}$, equal to $\sqrt{\sum_i A_i^2}$

θ the angle between the two vectors

Dividing by both lengths is what removes magnitude. The result lies in $[-1, +1]$, and for non-negative data such as counts it lies in $[0, 1]$:

$\cos\theta = 1$ identical direction, $\cos\theta = 0$ perpendicular, nothing in common, $\cos\theta = -1$ opposite.

Because algorithms usually expect a distance, cosine similarity is converted by

$$d_{\cos}(\mathbf{A}, \mathbf{B}) = 1 - \cos(\theta). \tag{8}$$

Worked example 4.1 — $\mathbf{A} = (1, 2)$ *and* $\mathbf{B} = (3, 6)$

Dot product

$$\mathbf{A}\cdot\mathbf{B} = (1)(3) + (2)(6) = 3 + 12 = 15$$

Length of each vector

$$\|\mathbf{A}\| = \sqrt{1^2 + 2^2} = \sqrt{5}, \qquad \|\mathbf{B}\| = \sqrt{3^2 + 6^2} = \sqrt{45}$$

Divide

$$\cos\theta = \frac{15}{\sqrt{5}\,\sqrt{45}} = \frac{15}{\sqrt{225}} = \frac{15}{15} = 1.000$$

The similarity is exactly 1 because $\mathbf{B} = 3\mathbf{A}$: the two vectors point the same way and differ only in length. For comparison, the Euclidean distance between them is $\sqrt{(1-3)^2 + (2-6)^2} = \sqrt{20} \approx 4.47$, which would call them quite different.

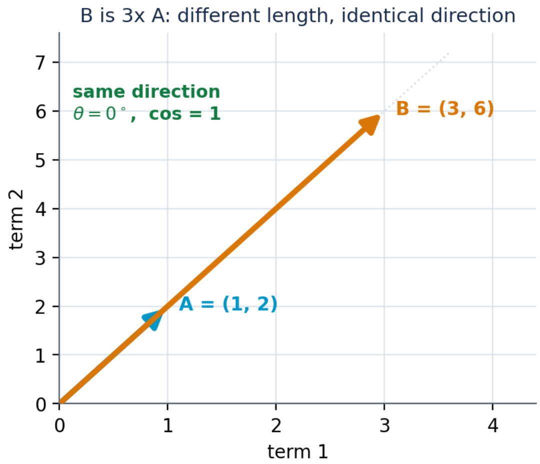

Figure 3: **B** is three times **A**. The angle between them is zero, so the cosine similarity is 1.

Worked example 4.2 — *Two documents with little in common*

Word counts over four terms: $\mathbf{A} = (2, 1, 0, 3)$ and $\mathbf{B} = (0, 3, 4, 0)$.

Dot product

$$(2)(0) + (1)(3) + (0)(4) + (3)(0) = 3$$

Lengths

$$\|\mathbf{A}\| = \sqrt{4+1+0+9} = \sqrt{14} \approx 3.742, \qquad \|\mathbf{B}\| = \sqrt{0+9+16+0} = \sqrt{25} = 5$$

Divide

$$\cos\theta = \frac{3}{3.742 \times 5} = \frac{3}{18.708} \approx 0.160$$

Close to zero: these two documents share very little.

5 Distance and similarity matrices

A **distance matrix** D records the distance between every pair of objects, so $D_{ij} = d(x_i, x_j)$ and D is $n \times n$. Three properties reduce the work:

- the diagonal is zero, since $d(x_i, x_i) = 0$;
- the matrix is symmetric, since d is symmetric;
- therefore only the $\frac{n(n-1)}{2}$ entries above the diagonal need computing — for $n = 4$ that is six calculations, not sixteen.

A **similarity matrix** is the same construction using a similarity measure. Its diagonal is 1, because every object is perfectly similar to itself.

5.1 A worked distance matrix

Take four points:

$$A = (2, 5), \qquad B = (5, 9), \qquad C = (8, 5), \qquad D = (3, 6).$$

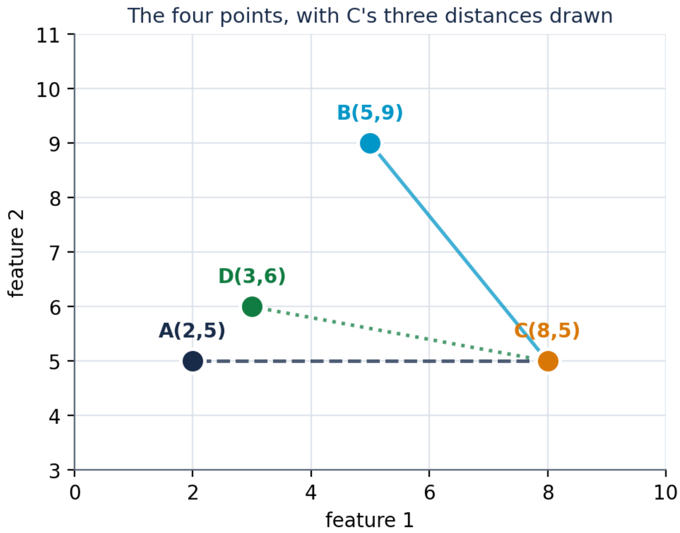

Figure 4: The four points. The three lines show C's distances to B, A and D.

Worked example 5.1 — *All six Euclidean distances*

$$d(A, B) = \sqrt{(2 - 5)^2 + (5 - 9)^2} = \sqrt{9 + 16} = 5.000$$
$$d(A, C) = \sqrt{(2 - 8)^2 + (5 - 5)^2} = \sqrt{36 + 0} = 6.000$$
$$d(A, D) = \sqrt{(2 - 3)^2 + (5 - 6)^2} = \sqrt{1 + 1} \approx 1.414$$
$$d(B, C) = \sqrt{(5 - 8)^2 + (9 - 5)^2} = \sqrt{9 + 16} = 5.000$$
$$d(B, D) = \sqrt{(5 - 3)^2 + (9 - 6)^2} = \sqrt{4 + 9} \approx 3.606$$
$$d(C, D) = \sqrt{(8 - 3)^2 + (5 - 6)^2} = \sqrt{25 + 1} \approx 5.099$$

Worked example 5.2 — *All six Manhattan distances*

$$d(A, B) = |2 - 5| + |5 - 9| = 3 + 4 = 7 \qquad d(B, C) = |5 - 8| + |9 - 5| = 3 + 4 = 7$$
$$d(A, C) = |2 - 8| + |5 - 5| = 6 + 0 = 6 \qquad d(B, D) = |5 - 3| + |9 - 6| = 2 + 3 = 5$$
$$d(A, D) = |2 - 3| + |5 - 6| = 1 + 1 = 2 \qquad d(C, D) = |8 - 3| + |5 - 6| = 5 + 1 = 6$$

Euclidean

	A	B	C	D
A	0.000	5.000	6.000	1.414
B	5.000	0.000	5.000	3.606
C	6.000	5.000	0.000	5.099
D	1.414	3.606	5.099	0.000

Manhattan

	A	B	C	D
A	0	7	6	2
B	7	0	7	5
C	6	7	0	6
D	2	5	6	0

Table 1: The same four points under the two metrics.

Read row C in both tables. Under the Euclidean metric, C's nearest neighbour is B at 5.000, ahead of D at 5.099 and A at 6.000. Under the Manhattan metric, B is C's *furthest* neighbour at 7, while A and D are both at 6. The same point moves from nearest to furthest purely because the metric changed.

5.2 A worked similarity matrix

Three documents, as counts over three terms:

$$d_1 = (2, 1, 0), \qquad d_2 = (4, 2, 0), \qquad d_3 = (0, 1, 3).$$

Worked example 5.3 — *Cosine similarities*

Lengths, computed once and reused

$$\|d_1\| = \sqrt{5} \approx 2.236, \quad \|d_2\| = \sqrt{20} \approx 4.472, \quad \|d_3\| = \sqrt{10} \approx 3.162$$

Each pair

$$S_{12} = \frac{(2)(4) + (1)(2) + (0)(0)}{\sqrt{5}\sqrt{20}} = \frac{10}{\sqrt{100}} = 1.000$$
$$S_{13} = \frac{(2)(0) + (1)(1) + (0)(3)}{\sqrt{5}\sqrt{10}} = \frac{1}{\sqrt{50}} \approx 0.141$$
$$S_{23} = \frac{(4)(0) + (2)(1) + (0)(3)}{\sqrt{20}\sqrt{10}} = \frac{2}{\sqrt{200}} \approx 0.141$$

$S_{12} = 1$ exactly, because $d_2 = 2d_1$: twice as many words, identical proportions. S_{13} and S_{23} are equal for the same reason.

	d_1	d_2	d_3
d_1	1.000	1.000	0.141
d_2	1.000	1.000	0.141
d_3	0.141	0.141	1.000

Table 2: Cosine similarity matrix. The diagonal is 1, not 0.

6 When and why the choice of distance matters

Four situations decide which metric is appropriate. In each, choosing wrongly does not raise an error — it silently produces a different answer.

6.1 Feature scale

Euclidean distance adds squared differences, so a feature measured over a large numerical range contributes far more than one measured over a small range, regardless of which is more important.

Worked example 6.1 — *Age and income*

Three customers, described by age in years and income in rupees:

$$P = (25,\ 300\,000), \qquad Q = (26,\ 900\,000), \qquad R = (55,\ 320\,000).$$

Unscaled Euclidean distances

$$d(P,Q) = \sqrt{(25-26)^2 + (300\,000 - 900\,000)^2} = \sqrt{1 + 3.6 \times 10^{11}} \approx 600\,000$$
$$d(P,R) = \sqrt{(25-55)^2 + (300\,000 - 320\,000)^2} = \sqrt{900 + 4 \times 10^{8}} \approx 20\,000$$

R is P's nearest neighbour by a factor of about 30. A thirty-year age gap contributed 900 to the sum; a twenty-thousand-rupee income gap contributed 400 000 000. Age is present in the calculation and has no influence on it.

Standardise each feature

Using $z = (x - \mu)/\sigma$ with $\mu_{\text{age}} = 40$, $\sigma_{\text{age}} = 15$, $\mu_{\text{inc}} = 500\,000$ and $\sigma_{\text{inc}} = 300\,000$:

$$P \to (-1.000,\ -0.667), \qquad Q \to (-0.933,\ +1.333), \qquad R \to (+1.000,\ -0.600)$$

Scaled Euclidean distances

$$d(P,Q) = \sqrt{(-1.000 + 0.933)^2 + (-0.667 - 1.333)^2} = \sqrt{0.004 + 4.000} \approx 2.001$$
$$d(P,R) = \sqrt{(-1.000 - 1.000)^2 + (-0.667 + 0.600)^2} = \sqrt{4.000 + 0.004} \approx 2.001$$

The two distances are now equal, which is the correct result: P differs from Q by two standard deviations of income and from R by two standard deviations of age.

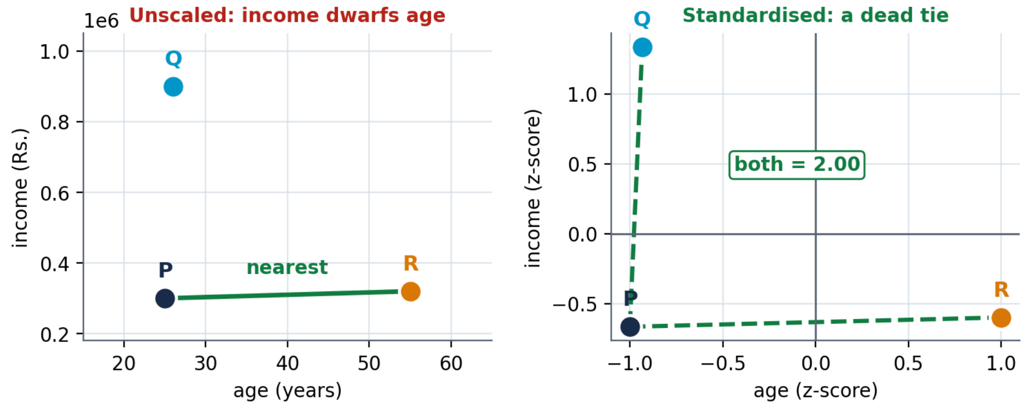

Figure 5: The same three customers before and after standardisation. Only the units changed.

The standardisation formula is

$$z = \frac{x - \mu}{\sigma}, \tag{9}$$

after which every feature has mean 0 and standard deviation 1. A z-score is also dimensionless, which is the only reason it is legitimate to add contributions from features measured in different physical units.

6.2 Magnitude against direction

If the overall size of a vector is irrelevant — the length of a document, the total spend of a customer, how generously a user rates films — then Euclidean distance will separate objects that should be treated as similar. Cosine similarity removes magnitude and compares proportions instead.

The converse also holds. If magnitude *is* the signal, cosine is the wrong choice: a customer spending Rs. 100 and one spending Rs. 100 000 in the same proportions have a cosine similarity of exactly 1.

6.3 Concentrated against spread-out differences

Because Euclidean distance squares each difference, it penalises a single large gap more heavily than several small ones. Manhattan distance, which adds the differences, does the opposite. This is why the two metrics can disagree about which point is nearest, as they did for point C above. Manhattan is also more robust when a feature contains occasional extreme values.

6.4 Number of features

As the number of features grows, the distances between all pairs of points become increasingly similar to one another. The contrast between the nearest and the furthest neighbour shrinks, and “nearest neighbour” stops being a meaningful idea. Adding an irrelevant feature is therefore not free: it dilutes every distance you compute. The usual responses are to select fewer, better features, to reduce dimensionality, or to use cosine similarity, which degrades more gracefully on sparse high-dimensional data.

If the data is...	**Use**	**Because**
continuous, comparable units	Euclidean	straight-line closeness is meaningful
movement restricted to axes; count features	Manhattan	it measures the distance actually travelled
contains occasional extreme values	Manhattan	no squaring, so one large gap cannot dominate
text, ratings, sparse counts	Cosine	length is noise; proportion is the signal
features on very different ranges	*standardise first*	no metric survives unscaled features

Table 3: Choosing a metric from the shape of the data.

7 The silhouette score

Once a clustering exists, its quality must be judged without any ground truth. The silhouette score does this geometrically: it asks whether each point is close to its own cluster and far from the others.

For a single point i:

$$s(i) \;=\; \frac{b(i) - a(i)}{\max\{a(i),\, b(i)\}} \tag{10}$$

where

$a(i)$ the mean distance from i to every *other* point in its own cluster — how well it fits where it is

$b(i)$ the mean distance from i to all points of the *nearest other* cluster — how well it would fit next door

max dividing by the larger of the two forces $-1 \le s(i) \le +1$

Reading the value:

- $s(i)$ near $+1$: $b \gg a$, so the point sits well inside its cluster and far from the next. Well clustered.
- $s(i)$ near 0: $a \approx b$, so the point lies on the boundary between two clusters. Ambiguous.
- $s(i)$ negative: $a > b$, so the point is on average closer to another cluster than to its own. Probably misassigned.

The mean of $s(i)$ over all points scores the whole clustering. A point that is alone in its cluster is assigned $s = 0$ by convention, because $a(i)$ is then undefined.

7.1 A well-separated clustering

$$C_1 = \{P_1(1,1),\ P_2(2,1),\ P_3(1,2)\}, \qquad C_2 = \{P_4(6,6),\ P_5(7,6),\ P_6(6,7)\}$$

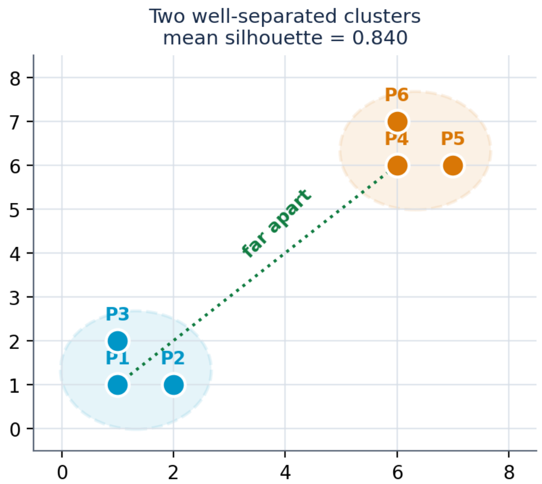

Figure 6: Two compact, well-separated clusters.

Worked example 7.1 — *Computing $s(P_1)$*

$a(P_1)$: mean distance to the other members of C_1

$$d(P_1, P_2) = 1, \qquad d(P_1, P_3) = 1 \qquad \Longrightarrow \qquad a(P_1) = \frac{1+1}{2} = 1.000$$

$b(P_1)$: mean distance to every member of C_2

$$d(P_1, P_4) = \sqrt{25+25} = \sqrt{50} \approx 7.071$$
$$d(P_1, P_5) = \sqrt{36+25} = \sqrt{61} \approx 7.810$$
$$d(P_1, P_6) = \sqrt{25+36} = \sqrt{61} \approx 7.810$$

$$b(P_1) = \frac{7.071 + 7.810 + 7.810}{3} \approx 7.564$$

Apply the formula

$$s(P_1) = \frac{7.564 - 1.000}{\max\{1.000,\ 7.564\}} = \frac{6.564}{7.564} \approx 0.868$$

Point	Cluster	$a(i)$	$b(i)$	$s(i)$
$P_1(1,1)$	C_1	1.000	7.564	0.868
$P_2(2,1)$	C_1	1.207	6.895	0.825
$P_3(1,2)$	C_1	1.207	6.895	0.825
$P_4(6,6)$	C_2	1.000	6.626	0.849
$P_5(7,6)$	C_2	1.207	7.364	0.836
$P_6(6,7)$	C_2	1.207	7.364	0.836
		Mean silhouette		**0.840**

Table 4: Every point scores highly, and the mean is close to 1.

7.2 A clustering with a misassigned point

$$Q_1(1,1),\quad Q_2(2,2),\quad Q_3(8,8),\quad Q_4(9,9),\qquad C_1=\{Q_1,Q_2,Q_3\},\quad C_2=\{Q_4\}$$

Q_3 plainly belongs with Q_4. The score detects this.

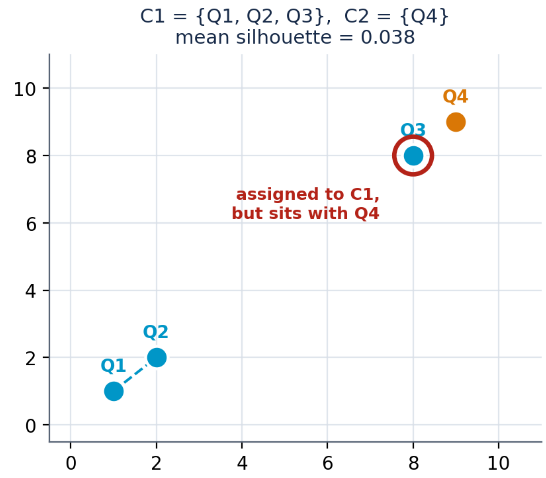

Figure 7: Q_3 has been placed in C_1 but lies beside Q_4.

Worked example 7.2 — *Computing $s(Q_3)$*

Every distance here is a multiple of $\sqrt{2}$, so the arithmetic stays exact.

$a(Q_3)$

$$d(Q_3,Q_1)=\sqrt{49+49}=7\sqrt{2},\qquad d(Q_3,Q_2)=\sqrt{36+36}=6\sqrt{2}$$

$$a(Q_3)=\frac{7\sqrt{2}+6\sqrt{2}}{2}=6.5\sqrt{2}\approx 9.192$$

$b(Q_3)$, the distance to the only other cluster

$$b(Q_3)=d(Q_3,Q_4)=\sqrt{1+1}=\sqrt{2}\approx 1.414$$

Apply the formula

$$s(Q_3)=\frac{\sqrt{2}-6.5\sqrt{2}}{6.5\sqrt{2}}=-\frac{5.5}{6.5}=-\frac{11}{13}\approx -0.846$$

The value is strongly negative: Q_3 is roughly six times closer to the other cluster than to its own.

Point	Cluster	$a(i)$	$b(i)$	$s(i)$
$Q_1(1,1)$	C_1	$4\sqrt{2} \approx 5.657$	$8\sqrt{2} \approx 11.314$	$+0.500$
$Q_2(2,2)$	C_1	$3.5\sqrt{2} \approx 4.950$	$7\sqrt{2} \approx 9.899$	$+0.500$
$Q_3(8,8)$	C_1	$6.5\sqrt{2} \approx 9.192$	$\sqrt{2} \approx 1.414$	-0.846
$Q_4(9,9)$	C_2	alone in its cluster		0.000
			Mean silhouette	**0.038**

Table 5: One badly placed point drags the mean from 0.840 down to 0.038.

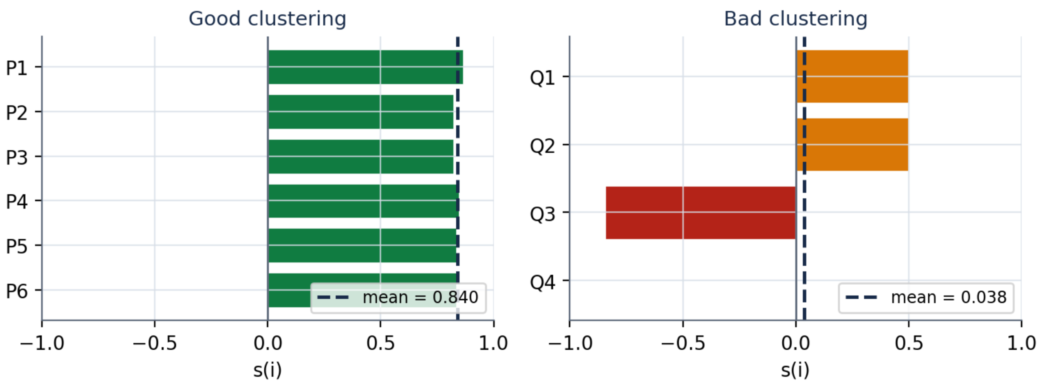

Figure 8: Per-point silhouettes for both clusterings.

Two cautions. First, always inspect the individual scores rather than the mean alone: in the second example two points scored a respectable $+0.5$ while one scored -0.846. Every negative value marks a point the algorithm has misplaced. Second, the silhouette score measures geometry only. A clustering of entirely structureless data can still return a positive score, so a high value does not establish that the groups are meaningful.

8 Numerical questions

Work these by hand and show every step.

1. For $p = (3, 4)$ and $q = (7, 1)$, compute $d_{\text{euc}}(p, q)$ and $d_{\text{man}}(p, q)$ using

$$d_{\text{euc}} = \sqrt{\sum_i (p_i - q_i)^2}, \qquad d_{\text{man}} = \sum_i |p_i - q_i|.$$

   Confirm that $d_{\text{man}} \geq d_{\text{euc}}$.

2. For $\mathbf{u} = (1, 0, 2)$ and $\mathbf{v} = (2, 0, 4)$, compute the cosine similarity. State what the result says about the two vectors.

3. A student reports $d_{\text{euc}} = 8.2$ and $d_{\text{man}} = 6.5$ for the same pair of points. Explain, without knowing the points, why this must be an error.

4. For $X = (0, 0)$, $Y = (3, 4)$ and $Z = (6, 0)$, construct the 3×3 Euclidean distance matrix.

5. For the same three points, construct the 3×3 Manhattan distance matrix. Then state which point is X's nearest neighbour under each metric, and whether the two agree.

6. Construct the 3×3 cosine similarity matrix for $d_1 = (1, 0, 1)$, $d_2 = (2, 0, 2)$ and $d_3 = (0, 1, 0)$. Interpret the value of S_{13}.

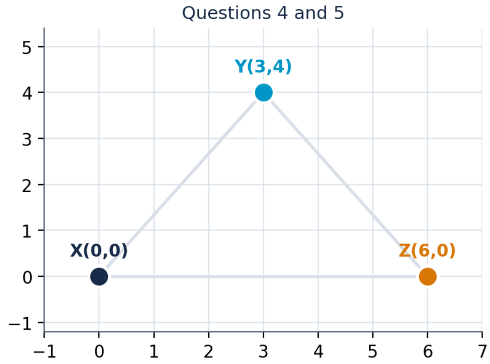

Figure 9: The three points used in questions 4 and 5.

7. Two points differ by 0.8 in feature A, whose range is 0 to 1, and by 10 in feature B, whose range is 0 to 1000. Compute each feature's contribution to the squared Euclidean distance, and the percentage contributed by B.

8. For the values $\{2, 4, 4, 4, 5, 5, 7, 9\}$, compute the mean, the population standard deviation, and the z-score of the value 9 using $z = (x - \mu)/\sigma$.

9. Cluster $A = \{(0, 0),\ (0, 2)\}$ and cluster $B = \{(6, 0),\ (6, 2)\}$. Compute a, b and s for the point $(0, 0)$ using $s(i) = \big(b(i) - a(i)\big)/\max\{a(i), b(i)\}$.

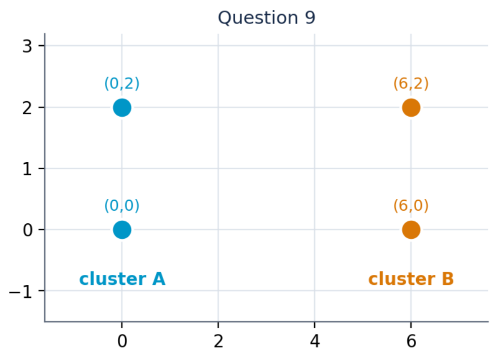

Figure 10: The two clusters used in question 9.

10. A point has $a(i) = 4.0$ and $b(i) = 2.0$. Compute $s(i)$ and state what it implies.

11. Using the four points $A(2, 5)$, $B(5, 9)$, $C(8, 5)$, $D(3, 6)$ and the Euclidean distance matrix computed earlier, suppose the clustering is $C_1 = \{A, D\}$ and $C_2 = \{B, C\}$. Compute $s(A)$.

12. Two customers are described by their spending across five product categories: $\mathbf{u} = (2, 1, 4, 0, 1)$ and $\mathbf{v} = (20, 10, 40, 0, 10)$. Compute both the Euclidean distance and the cosine similarity. Explain why the two measures disagree so strongly, and state which you would use to group customers by *taste* rather than by *budget*.